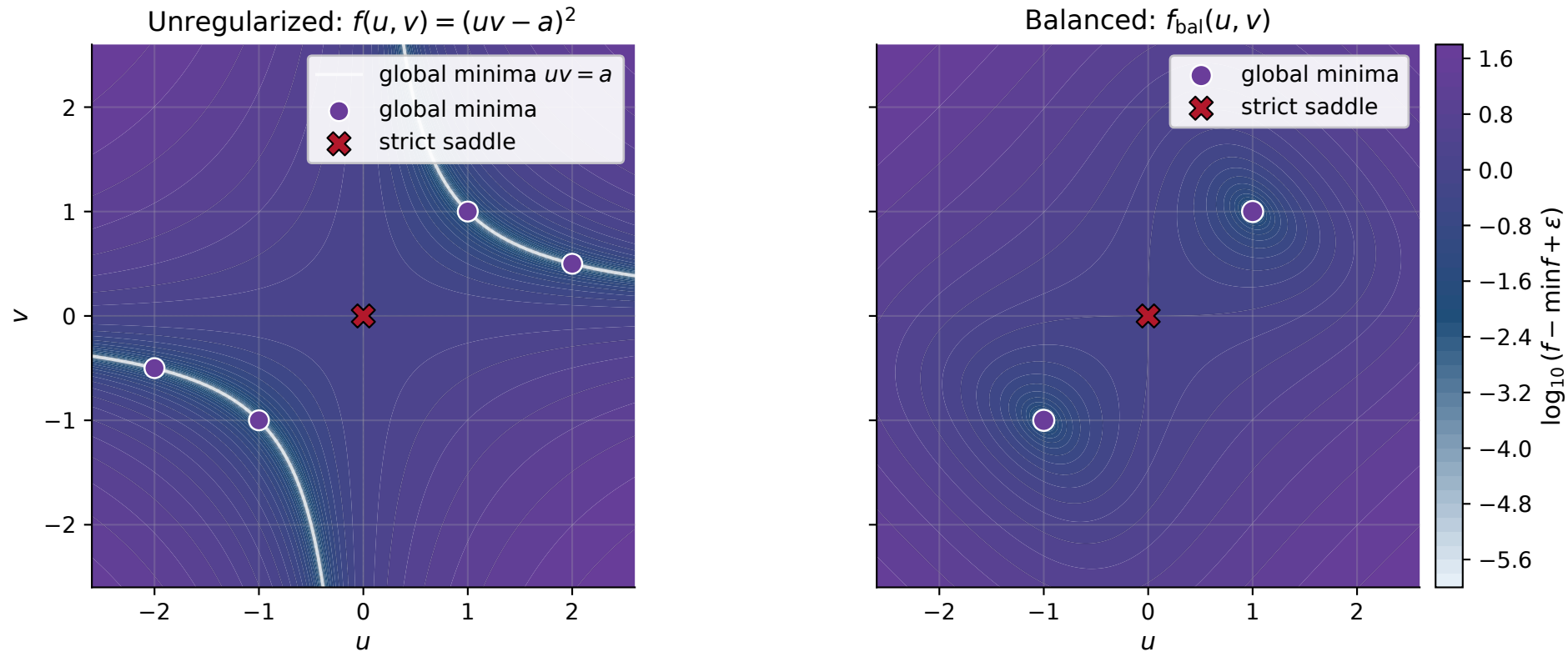
Unregularized: $f(u,v) = (uv - a)^2$
global minima $uv = a$
global minima
strict saddle
v
u
Balanced: $f_{\text{bal}}(u,v)$
global minima
strict saddle
u
$\log_{10}(f - \min f + \varepsilon)$